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AI-Assisted Recruitment Decision Support Platform

A Product Case Study — Nishat Zerín

The One-Line Pitch

An AI system that ranks and explains candidate CVs for recruiters — but never auto-rejects, never penalizes CV formatting, and never lets application timing affect the outcome. Built as a working demonstration of how to govern AI output, not just call an API.

1. The Problem

- Recruiters get ~500 CVs per role and spend most of their time on repetitive first-pass screening, not high-judgment work
- Manual screening is slow and inconsistent between reviewers
- Legacy ATS keyword tools penalize CVs with non-standard formatting, such that, good candidates get filtered out for reasons unrelated to skill
- A subtler bias: recruiters unconsciously favor early applicants over later (even stronger) ones
- **Regulatory context:** the EU AI Act classifies recruitment AI as high-risk, requiring human oversight and auditability and this isn't a hypothetical concern

2. Why This Is the Headline Feature: Governing AI, Not Just Using It

Most “AI screening” tools are a thin wrapper around an LLM call. This platform is built around a different idea: **the AI proposes, a human decides, and every step in between is traceable.**

Two things make this concrete, not just a claim:

1. Grounded explanations, not free-form narration. Every recommendation is built from specific extracted facts (“matched 4/6 required skills, missing X”). Not only an open-ended LLM paragraph that sounds plausible but might not be accurate.

2. A real agentic workflow with a human checkpoint.

When a recruiter asks “*Why does Candidate 7 need manual review?*”, the system runs a defined chain:

retrieve extracted data → check extraction confidence → check missing requirements → retrieve the relevant policy → generate a grounded recommendation (Review / Proceed / Need More Information).

If a step can’t retrieve what it needs, it says so and doesn’t fabricate an answer.

The system never auto-rejects. Every candidate stays visible to a human. This is AI as a working system with governance built in with controlled inputs, traceable outputs, human review — not a chatbot bolted onto a workflow.

3. Key Product Decisions (Where the PO Judgment Shows)

Four stakeholders want different things, and the design has to hold all of them at once:

Stakeholder	Wants	Tension
HR Manager	Faster screening	vs. quality and fairness
Hiring Manager	A short, high-quality shortlist	vs. speed
Recruiter	Trust and explainability	vs. simplicity
Candidate	A fair shot regardless of formatting or timing	vs. automation efficiency

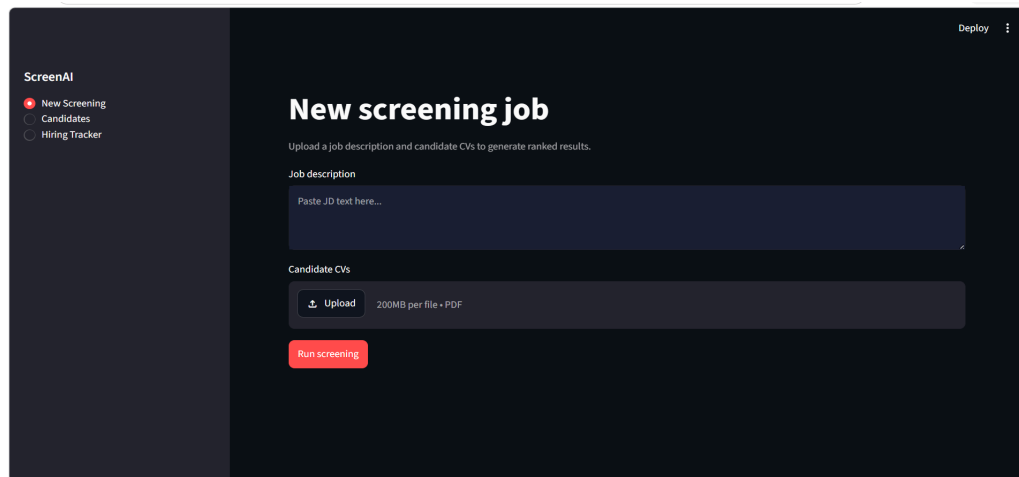
Two design decisions resolve this tension directly:

- **Dual-signal scoring** — extraction confidence is tracked *separately* from match score. A CV that couldn’t be reliably parsed is never silently ranked low, instead it’s routed to manual review, regardless of its score. This turns “AI might penalize non-standard CVs” from a risk into a designed, testable safeguard.
- **Batch-only ranking** — candidates are only ranked once the full batch is in, purely by score. Submission timestamp is logged for audit only, never used in scoring. A strong candidate applying near the deadline is judged exactly the same as one who applied on day one.

4. What Recruiters Actually See — Working Prototype

(This is no longer a wireframe — these are screenshots of the actual working Streamlit application, running on real (mocked) candidate data with a live LLM call behind “Why this rank?”)

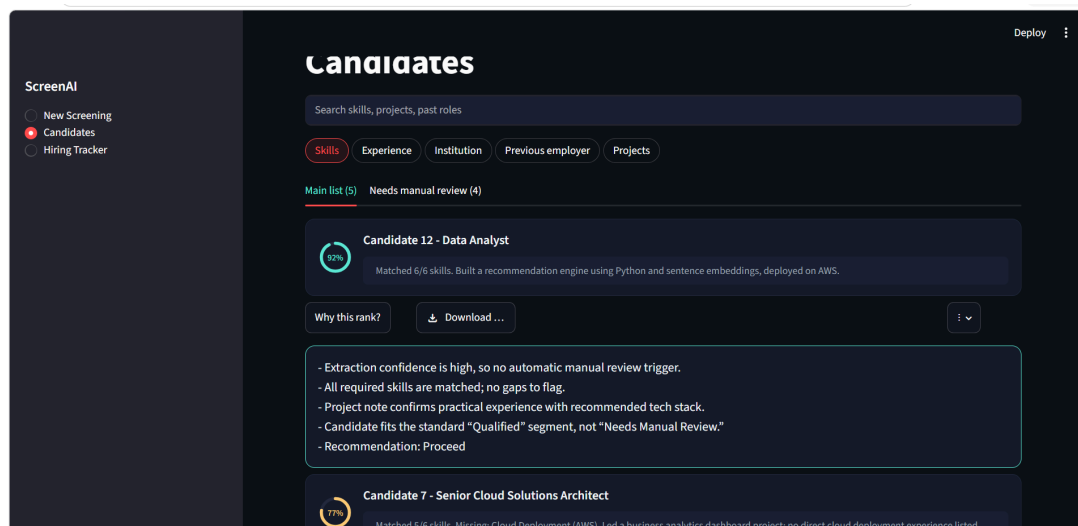
Upload screen:



The screenshot shows the 'New screening job' interface. On the left is a sidebar with 'ScreenAI' and three radio buttons: 'New Screening' (selected), 'Candidates', and 'Hiring Tracker'. The main area has a title 'New screening job' and a subtitle 'Upload a job description and candidate CVs to generate ranked results.' Below this is a 'Job description' section with a text area labeled 'Paste JD text here...'. Underneath is a 'Candidate CVs' section with an 'Upload' button and the text '200MB per file • PDF'. At the bottom is a red 'Run screening' button. A 'Deploy' link is in the top right corner.

New screening upload screen

Candidates screen — ranked list with the agentic explanation expanded:



The screenshot shows the 'Candidates' interface. The sidebar is the same as the previous screen. The main area has a title 'Candidates' and a search bar 'Search skills, projects, past roles'. Below the search bar are five tabs: 'Skills' (selected), 'Experience', 'Institution', 'Previous employer', and 'Projects'. Under the tabs, there are two sections: 'Main list (5)' and 'Needs manual review (4)'. The 'Main list (5)' section shows two candidate cards. The first card is for 'Candidate 12 - Data Analyst' with a 92% match. It includes a 'Why this rank?' button, a 'Download ...' button, and a dropdown menu. The explanation for this candidate is: '- Extraction confidence is high, so no automatic manual review trigger.', '- All required skills are matched; no gaps to flag.', '- Project note confirms practical experience with recommended tech stack.', '- Candidate fits the standard “Qualified” segment, not “Needs Manual Review.”', and '- Recommendation: Proceed'. The second card is for 'Candidate 7 - Senior Cloud Solutions Architect' with a 77% match. The explanation for this candidate is: 'Matched 5/6 skills. Missing: Cloud Deployment (AWS). Led a business analytics dashboard project; no direct cloud deployment experience listed.'

Candidates screen with explanation

Notice the explanation isn't a generic paragraph — it's grounded, bulleted, and ends with an explicit recommendation, exactly matching the agentic chain described in Section 2.

A third screen (Hiring Tracker) was also built during implementation but isn't pictured here. It's a Kanban-style board (Shortlisted → Interview → Hired) showing each

candidate's contact info and a "View original CV" action, useful once a candidate moves past initial screening.

What's real vs. what's UI-only in this build, stated plainly:

Real: search, tab filtering, candidate ranking by match score, the agentic explanation (a live call to an open-weight LLM via Groq), and the show/hide toggle on explanations

UI-only (designed, not wired): all 5 filter pills (Skills, Experience, Institution, Previous employer, Projects) are currently cosmetic — none of them change the results shown; ranking is always by match score regardless of pill selection. The "Download CV" / "View original CV" buttons, the "Schedule interview / Reject / Compare with others" actions, and the tracker's "Move back a stage" action are all placeholders with no file or logic behind them. The "Run screening" button does not read the uploaded JD text or CV files. It always shows the same pre-loaded mock candidates regardless of what's uploaded.

The build prioritized proving the ranking + agentic-explanation logic over building every surrounding workflow action, and that tradeoff is disclosed here rather than left implicit.

5. Selected User Stories

- *As a recruiter*, I want ranked candidates with a clear explanation for each score, so I can trust and act on the recommendation quickly.
- *As a recruiter*, I want CVs that couldn't be reliably parsed flagged separately, so I don't overlook a strong candidate due to formatting.
- *As a recruiter*, I want to ask why a specific candidate needs manual review, so I get a traceable justification instead of digging through raw data myself.
- *As a candidate*, I want assurance that a formatting quirk in my CV won't cause me to be silently filtered out without human review.
- *As a compliance stakeholder*, I want every recommendation traceable to extracted facts and a documented rule, so decisions can be audited and defended.

(Full list of 20 user stories + acceptance criteria in the detailed BRD)

6. Competitive Landscape

Approach	Speed	Consistency	Fairness	Auditability
Manual screening	Slow	Low — varies by reviewer	Medium — human bias still present	None — no structured record
Legacy ATS (Greenhouse, Lever, keyword-based tools)	Fast	High	Low — formatting-brittle, penalizes non-standard CVs	Low — rejects with no explanation
Ad hoc LLM use (recruiter pastes CV into ChatGPT)	Fast	Low — no structure	Low — no safeguards	None — no audit trail

Approach	Speed	Consistency	Fairness	Auditability
		between uses		
This platform	Fast	High	High — dual-signal design catches formatting bias	High — every score traceable to extracted facts and rule

The market gap this closes: established ATS platforms solved speed and consistency years ago, but they still treat formatting-brittleness and unexplained rejection as acceptable trade-offs. Ad hoc LLM use solves flexibility but adds no structure at all. Nobody in the current market combines all four properties i.e speed, consistency, fairness, and auditability, in one system. That combination, not any single feature, is the actual gap.

Why this matters commercially, not just technically: the EU AI Act's high-risk classification for recruitment AI means the fairness/auditability gap isn't a nice-to-have. Vendors without a real answer to "how do you prevent formatting bias" or "can you show me why this candidate was ranked here" face growing regulatory exposure. This platform's core design *is* the answer to those two questions.

7. Prioritization (MoSCoW)

Priority	Requirements	Why
Must	FR-1 to FR-11 (core extraction, dual-signal scoring, review-routing, grounded explanations, no-auto-reject, audit logging)	The core value proposition doesn't exist without these — this is the platform
Must	FR-16 (agentic explanation workflow), FR-18 (synonym-aware matching), FR-20 (two-segment interface)	These directly demonstrate the platform's core claims such as AI governance, fairness, and visible safeguards. Without them, the differentiators are just described, not shown
Should	FR-12, FR-13 (skill filtering, adjustable weighting), FR-17 (deep project analysis), FR-23 (keyword search)	Meaningfully strengthen the product, and are realistic to build within the MVP window, but the platform still works without them
Could	FR-19 (extended filters: institution, employer, referral), FR-21 (per-candidate PDF download), FR-22 (recruiter	Genuinely useful recruiter workflow features; included in requirements and the wireframe, but reasonable to defer past MVP without weakening the core pitch

Priority	Requirements	Why
	notes), FR-25 (shortlist export)	
Won't (this round)	FR-24 (bulk shortlist), side-by-side candidate comparison	Real value, but lowest urgency relative to everything above such as explicitly backlog, not overlooked

What actually gets built vs. only specified: Must-have items are built end-to-end in the working demo. Most Should and Could items are fully specified here and shown in the wireframe, but only a couple (search and the two-segment toggle) are implemented in working code. A deliberate scope call given the build window, not a gap in the thinking.

8. Go-to-Market

Who this is for: - Mid-size companies (roughly 200-2,000 employees) with an internal recruitment team that is large enough to feel the ~500-CVs-per-role volume problem, but without the budget or need for an enterprise ATS overhaul. Recruitment teams already frustrated with a legacy ATS's keyword-matching limitations, or currently doing ad hoc, unstructured LLM use with no safeguards

How adoption would start (not a company-wide rollout on day one): - Pilot with a single hiring team on a single high-volume role, run alongside their existing process (not replacing it yet). Then the AI's recommendations are compared against what the team decided manually. Success in the pilot (faster shortlisting, no missed strong candidates in the review bucket) becomes the case for expanding to more roles/teams

How it competes for attention: - Positioned against legacy ATS tools as *"the fair, explainable alternative,"* directly targeting the two weaknesses named in the competitive analysis: formatting bias and no auditability. Positioned against ad hoc LLM use as *"the safe way to actually use AI for this"* for teams already experimenting informally but worried about consistency or compliance. The EU AI Act angle becomes a genuine sales argument, not just a risk disclosure: compliance-conscious buyers (especially in regulated markets) have a concrete reason to prefer a system built with human-in-the-loop auditability from day one

9. Roadmap

Now (MVP): structured extraction → dual-signal scoring → agentic explanation workflow → basic Streamlit UI → local logging for audit

Next: real ATS integration, expanded bias auditing, candidate-facing transparency, role-based access

Later: multi-role comparison, full analytics dashboard, and recruiter override tracking to improve matching over time

10. Success Metrics

- Screening time reduction (illustrative, clearly labeled as projected)
- Review-bucket rate (are low-confidence CVs being caught and are they being reviewed, not ignored)
- Ranking consistency across repeated runs
- Explanation groundedness (no fabricated details)
- Bias swap-test outcome (identical CV, varied name/university → no score difference)
- Order-independence check (identical CV, different timestamp → no score difference)

Full detailed requirements, all 25+ functional requirements, complete user story set, and acceptance criteria: see the accompanying Business Requirements & Solution Design Document.